QUANT

Mensuration & Geometry

(Triangles, Polygons & Mensuration)

Assignment 03

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Geometry Online Assignment Cluster I -03

1. The measure of an angle which is four times its complement is

1. 36° 2. 72° 3. 98° 4. 108°

2. The sum of the internal angles of a regular polygon of n sides, when expressed in right angles, is

1. $(2n - 2)$ right angles
2. $(n - 2)$ right angles
3. $(2n - 4)$ right angles
4. $(n - 4)$ right angles

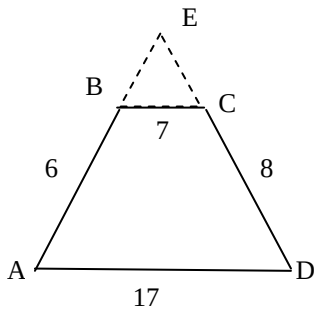
3. ABCD is a trapezium in which AB is parallel to DC and AB = 2CD. If its diagonals intersect each other at O, then the ratio of the areas of triangles AOB and COD is

1. 4 : 1 2. 1 : 2 3. 2 : 3 4. 4 : 5

4. What is the area of a regular octagon of side a?

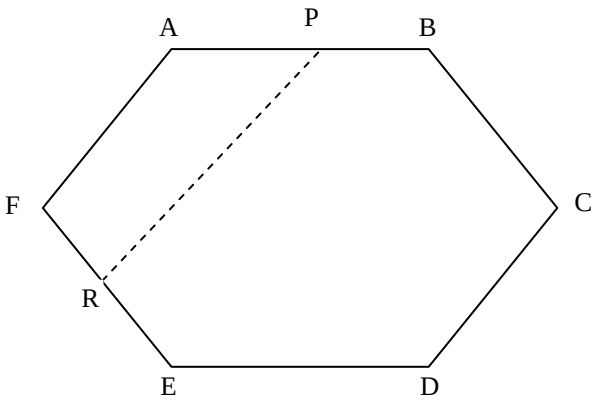
1. $(2 + 2\sqrt{2}) \times a^2$ 2. $(3 + \sqrt{2}) \times a^2$ 3. $(4 + \sqrt{2}) \times a^2$ 4. $(5 + \sqrt{2}) \times a^2$

5. In the trapezium ABCD shown below, AD is parallel to BC. AB = 6, BC = 7, CD = 8, AD = 17. If the sides AB and CD are extended to meet at E, find the measure of ∠ AED.



1. 60^0 2. 90^0 3. 100^0 4. 120^0

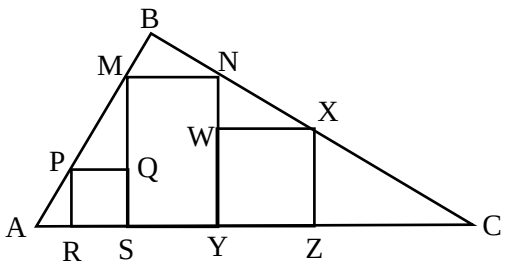
6.



ABCDEF is a regular hexagon AP : PB = 5 : 2 and PR is parallel to AF. Find the ratio of lengths of AF to PR.

1. $\frac{2}{7}$ 2. $\frac{5}{7}$ 3. $\frac{7}{12}$ 4. $\frac{2}{5}$

7.



In Δ ABC, ∠ B = 90°. □ PQRS is a square of side 5 units and □ WXYZ is a square of side 10 units. MS and SY are the length and breadth respectively of rectangle MNSY. What is the length of MS?

1. 12 cm 2. 15 cm 3. 25 cm 4. 20 cm

8. Find each internal angle of a regular polygon of 10 sides .

1. 144° 2. 72° 3. 98° 4. 108°

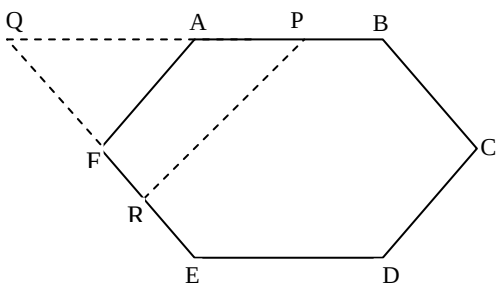
9. If each internal angle of a regular polygon is 140^0. Find its side.

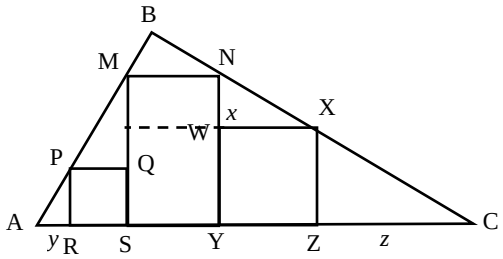
1. 10 2. 9 3. 12 4. None of these

10. If each external angle of a regular polygon is 18^0. Find its side.

1. 10 2. 9 3. 12 4. None of these

Answer key & Explanations

Q. No.	Ans.	Exp.
1.	2	Let the angle be x. $\therefore x = 4(90 - x)$ $\Rightarrow x = 360 - 4x$ $\Rightarrow 5x = 360 \Rightarrow x = 72$.
2.	3	Sum of internal angles of a regular polygon of n sides $= (2n - 4)$ right angles.
3.	1	$AB \parallel CD$. $AB = 2CD$. This is an isosceles trapezium. When diagonals intersect at O, triangles formed are AOB and COD. These 2 triangles are similar to each other. Hence ratio of their areas = ratio of $(\text{sides})^2$ $= (2 : 1)^2$ $= 4 : 1$
4.	1	The area of the octagon can be expressed as the difference of the larger square that comprises the octagon and the 4 right triangles at the corners that we can deduct. Each of the right triangles will have the two perpendicular sides as $a / \sqrt{2}$. The side of the bigger square will be $a + a/\sqrt{2} + a/\sqrt{2} = (1 + \sqrt{2})\, a$. Squaring we get the area of the square as $(1 + 2 + 2\sqrt{2})a^2$ and the 4 triangles as $4 \times ½ \times (1/\sqrt{2})^2 \times a^2$. The difference between the two works out to $(2+2\sqrt{2}) \times a^2$.
5.	2	CF is drawn parallel to BA from C $\Rightarrow$ CF = BA = 6, FD = AD – AF = AD – BC = 17 – 7 = 10 In Δ CFD, CF = 6, FD = 10, CD = 8 $\Rightarrow$ CFD is right angled at $\angle$ C Thus $\angle$ FCD = 90^0 $\Rightarrow \angle$ AED = 90^0
6.	3	 Here Δ AQF is an equilateral triangle So AF = AQ But AF || PR. Then Δ AQF and Δ PQR are similar So $\frac{AQ}{PQ} = \frac{AF}{PR} \Rightarrow \frac{AF}{PR} = \frac{7}{7+5} = \frac{7}{12}$
7.	2	

		 Suppose WN = x, AR = y and ZC = z. Now, MQ = x + 5. Since the vertical and horizontal lines are parallel, we can use similarity. Δ MQP $\approx$ Δ MSA $\Rightarrow$ MQ / MS = PQ / AS $\Rightarrow$ (x + 5) / (x + 10) = 5 / (y + 5). Solving this gives $xy + 5y = 25 \rightarrow$ Eqn. (I). Δ CZX $\approx$ Δ CYN $\Rightarrow$ CZ / YC = XZ / NY $\Rightarrow$ z / (z + 10) = 10 / (x + 10). Solving this gives $xz = 100 \rightarrow$ Eqn. (II). Δ CZX $\approx$ Δ PRA $\Rightarrow$ CZ / XZ = PR / AR $\Rightarrow$ z / 10 = 5 / y. Solving this gives $yz = 50 \rightarrow$ Eqn. (III). From Eqns. (II) and (III), we get $xz = 100 = 2yz \Rightarrow x = 2y$. Substituting this value in Eqn. (I) we get $2y^2 + 5y - 25 = 0$. Solving this we get $y = 5/2$. So, $x = 5$. Thus, MS = x + 10 = 15 cm.
8.	1	Each angle= $\frac{(n-2)\times 180}{n}$ $= \frac{(10-2)\times 180}{10}$ $= 144^0$
9.	2	Each internal angle = 140^0 So external angle $=180^0 - 140^0 = 40^0$ Hence number of sides= $\frac{360}{40}$ =9
10.	4	Each external angle $=18^0$ Hence number of sides= $\frac{360}{18}$ =20